

Estructura de Datos Enriched - Ejemplos

📁 Cómo se Almacenan los Datos

1. Vocabulario Detallado con Pronunciación

```
{
  "categories": [
    {
      "name": "Sentimientos",
      "description": "Emotions related to happiness and well-being",
      "words": [
        {
          "word": "happiness",
          "translation": "felicidad",
          "pronunciation": "/ˈhæpɪnəs/",
          "example": "True happiness comes from within."
        },
        {
          "word": "anxiety",
          "translation": "ansiedad",
          "pronunciation": "/æŋˈzaɪəti/",
          "example": "Meditation reduces anxiety."
        }
      ]
    }
  ]
}
```

Campos Almacenados en BD:

- **word**: Palabra en inglés
- **definition**: Traducción + ejemplo combinados
- **example**: Ejemplo completo
- **pronunciation**: Formato IPA
- **category_id**: Referencia a la categoría
- **order**: Posición en la categoría

2. Ejercicios de Escritura Progresivos

```
{
  "exercises": [
    {
      "type": "writing",
      "title": "Past Habits Reflection",
      "instructions": "Write 5-7 sentences about what you used to do that
```

```
is different now.",
  "example": "I used to think money was the only way to happiness. Now
I realize connections matter more.",
  "min_words": 50,
  "difficulty": "beginner"
},
{
  "type": "writing",
  "title": "Internet Impact Analysis",
  "instructions": "Discuss how the internet has changed your brain and
thinking patterns.",
  "example": "The internet has changed how I think. I used to read
books, but now I scroll quickly.",
  "min_words": 70,
  "difficulty": "intermediate"
},
{
  "type": "writing",
  "title": "Intelligence Redefined",
  "instructions": "Write about what intelligence means beyond just
IQ.",
  "example": "Intelligence is more than test scores. Emotional
intelligence matters too.",
  "min_words": 100,
  "difficulty": "advanced"
}
]
}
```

Tabla WritingPractice:

id	unit_id	title	instructions
	example_text	difficulty	order
---	-----	-----	-----
	-----	-----	-----
1	7	Past Habits Reflection	Write 5-7 sentences about...
	I used to think money...	beginner	0
2	7	Internet Impact Analysis	Discuss how internet changed...
	The internet has changed..	intermediate	1
3	7	Intelligence Redefined	Write about intelligence...
	Intelligence is more...	advanced	2
...			

3. Diálogos Prácticos

```
{
  "dialogues": [
    {
```

```

    "title": "Coffee Shop - What Makes You Happy",
    "lines": [
      {
        "speaker": "A",
        "text": "I used to think happiness was about money and success."
      },
      {
        "speaker": "B",
        "text": "But now you realize simple things matter more, right?"
      },
      {
        "speaker": "A",
        "text": "Exactly! Real happiness comes from relationships and
growth."
      }
    ]
  }
]
}

```

Almacenado en UnitExtra como JSON:

```

{
  "unit_id": 7,
  "data": {
    "dialogues": [
      {
        "title": "Coffee Shop - What Makes You Happy",
        "lines": [...]
      }
    ]
  }
}

```

4. Reglas Gramaticales

```

{
  "grammar": [
    {
      "title": "Articles (a/an/the/no article)",
      "rule": "Use 'the' with specific things, use 'a/an' with general
things.",
      "example": "I go to school (as a student) vs I go to the school (the
building). The happiness I feel is real.",
      "level": "beginner"
    },
    {
      "title": "Used to",

```

```

    "rule": "Express habits or states in the past that no longer occur.",
    "example": "I used to feel anxious. She used to spend hours on social
media.",
    "level": "beginner"
  }
]
}

```

Tabla GrammarRule:

```

id | unit_id | topic | rule | example
| order
---|-----|-----|-----|-----
-----|-----
1 | 7 | Articles | Use 'the' with specific... | I go to
school... | 0
2 | 7 | Used to | Express habits in past... | I used
to feel anxious... | 1
...

```

Estructura de Base de Datos Completa

Tabla: Units

```

id | unit_number | title | description
---|-----|-----|-----
-----
7 | 7 | MIND (La Mente) | Exploring topics about
happiness, the...
8 | 8 | ART (Arte) | Discover art, music, and
creative...
9 | 9 | MONEY (Dinero) | Understanding money, finance,
and...
10 | 10 | SCIENCE AND TECHNOLOGY | Exploring technology,
innovation, and...
11 | 11 | NATURAL WORLD | Understanding nature,
environment, and...
12 | 12 | MEDIA | Understanding media,
information, and...

```

Tabla: VocabularyCategory

```

id | unit_id | category_name | order
---|-----|-----|-----
1 | 7 | Sentimientos | 0
2 | 7 | Phrasal Verbs | 1

```

3		8		Géneros de Música		0
...						

Tabla: VocabularyItem

id	category_id	word	definition	example
		pronunciation	order	
1	1	happiness	felicidad - True happiness..	True happiness comes from..
2	1	anxiety	ansiedad - Meditation...	Meditation reduces anxiety
...				

Tabla: GrammarRule

id	unit_id	topic	rule	example
				order
1	7	Articles	Use 'the' with specific	I go to school...
2	7	Used to	Express habits in past	I used to feel anxious...
...				

Tabla: WritingPractice

id	unit_id	title	instructions	
		example_text	difficulty	order
1	7	Past Habits Reflection	Write 5-7 sentences...	I used to think...
2	7	Internet Impact...	Discuss how internet...	The internet has...
3	7	Intelligence Redefined	Write about intelligence	Intelligence is...
...				

Tabla: UnitExtra (JSON Storage)

--

id	unit_id	data (JSON)
1	7	{ "dialogues": [{ "title": "Coffee Shop...", "lines": [...] }], "exercises_count": 3 }
2	8	{ "dialogues": [...], "exercises_count": 2 }
...		

Flujo de Datos en la Plataforma

Cuando Un Estudiante Accede a una Unidad:

1. CARGAR VOCABULARIO
 - └ SELECT * FROM vocabulary_items
 - WHERE category_id IN (
 - SELECT id FROM vocabulary_categories
 - WHERE unit_id = 7
 -)
 - └ Mostrar: palabra, traducción, pronunciación, ejemplo
2. CARGAR GRAMÁTICA
 - └ SELECT * FROM grammar_rules
 - WHERE unit_id = 7
 - └ Mostrar: title, rule, example, level
3. CARGAR DIÁLOGOS
 - └ SELECT data FROM unit_extra
 - WHERE unit_id = 7
 - └ Parse JSON dialogues
 - └ Mostrar: título, conversación
4. CARGAR EJERCICIOS
 - └ SELECT * FROM writing_practices
 - WHERE unit_id = 7
 - └ Mostrar: título, instrucciones, ejemplo, min_words

Cuando Un Estudiante Envía un Ejercicio de Escritura:

1. GUARDAR INTENTO
 - └ INSERT INTO user_writing_submissions

```
(user_id, writing_practice_id, text, submitted_at)
```

2. ANALIZAR TEXTO

- └ Extraer palabras claves de reglas gramaticales
- └ Verificar presencia en el texto
- └ Calcular puntuación (0-100)
- └ Generar mensajes de feedback

3. GUARDAR RESULTADO

- └ UPDATE user_writing_submissions
- └ SET feedback = '...', score = 85
- └ Mostrar al usuario:
 - Puntuación
 - Mensajes de feedback
 - Áreas de mejora

🔍 Ejemplos de Queries Útiles

Obtener Todo Contenido de una Unidad:

```
SELECT
  u.unit_number,
  u.title,
  gc.category_name,
  vi.word,
  vi.pronunciation,
  vi.example,
  gr.topic,
  wp.title as exercise_title,
  wp.difficulty
FROM units u
LEFT JOIN vocabulary_categories vc ON u.id = vc.unit_id
LEFT JOIN vocabulary_items vi ON vc.id = vi.category_id
LEFT JOIN grammar_rules gr ON u.id = gr.unit_id
LEFT JOIN writing_practices wp ON u.id = wp.unit_id
WHERE u.unit_number = 7
ORDER BY vc.order, vi.order, gr.order, wp.order;
```

Obtener Ejercicios por Nivel de Dificultad:

```
SELECT wp.title, wp.instructions, wp.example_text, wp.difficulty
FROM writing_practices wp
JOIN units u ON wp.unit_id = u.id
WHERE u.unit_number = 7
AND wp.difficulty = 'beginner'
ORDER BY wp.order;
```

Obtener Vocabulario de una Categoría:

```
SELECT vi.word, vi.pronunciation, vi.definition, vi.example
FROM vocabulary_items vi
JOIN vocabulary_categories vc ON vi.category_id = vc.id
WHERE vc.category_name = 'Sentimientos'
AND vc.unit_id = (SELECT id FROM units WHERE unit_number = 7)
ORDER BY vi.order;
```

Ventajas de esta Estructura

✅ **Flexible:** Fácil agregar más contenido ✅ **Escalable:** Soporta crecer el número de unidades ✅

Queryable: Búsquedas complejas son rápidas ✅ **JSON Storage:** Diálogos y datos complejos sin

normalización excesiva ✅ **Progresivo:** Diferentes niveles de dificultad ✅ **Relaciones:** Vinculación clara entre tablas

Total de Contenido Almacenado

- **60 palabras de vocabulario** (10 por unidad × 6 unidades)
- **13+ ejercicios de escritura** (3 en Unit 7, 2 en otras)
- **12+ diálogos** (2 en Unit 7, 1-2 en otras)
- **15+ reglas gramaticales** (2-3 por unidad)
- **6 quizzes** (1 por unidad)
- **Pronunciaciones IPA:** 60+

Total en BD: ~200+ registros base + JSON estructurado